

Strong and Weak Epistemic Operators in Assertions, Polar Questions, and Declarative Questions: An Experimental Study of German

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Abstract. This article investigates the distribution of the strong subjective epistemic operators *sicherlich*, *wohl*, *wahrscheinlich*, *müsste* and the weak subjective epistemic operators *vielleicht*, *möglicherweise*, *könnte* in polar questions and in declarative questions in German. The motivation for the study is the hypothesis that in polar questions, the perspective of subjective operators is shifted to the addressee (so-called “interrogative flip”), whereas in declarative questions it remains with the speaker. The article hypothesizes that under this assumption, weak operators should be rated better in polar questions, whereas strong operators should get a higher rating in declarative questions. This is confirmed by an experiment, which consisted in 280 participants in a within-subject rating of assertions, polar questions and declarative sentences with these subjective epistemic operators. The article motivates the guiding hypotheses by pragmatic considerations, and attempts to explain the slightly deviating behavior of the discourse particle *wohl*.

1 Introduction

It is a long-held opinion that in assertions, epistemic operators serve to fine-tune the level of commitment of the speaker to the communicated proposition. For example, Lyons (1977: 797) stated “Any utterance in which the speaker explicitly qualifies his commitment to the truth of a proposition expressed by the sentence he utters [...] is an epistemic modal, or modalized utterance”. Palmer (1986) considers the modification of speaker’s commitment as an essential function of epistemic and also evidential operators. Caffi (2013: 269), in a survey of mitigation devices, states “a whole class is made up of epistemic mitigators and evidentials, which are specialized for assertives”. Wolf (2015) develops a theoretical proposal in which the assertion of an epistemically weakened proposition amounts to a weaker assertion. Van Elswyk (2024) argues that epistemic expressions are hedges of assertions.

In Krifka (2026), I report on an investigation of assertions modified by epistemic and evidential operators, and presented experimental evidence that subjective epistemic operators generally weaken the commitment of the speaker to the proposition, which holds even for strong subjective operators like the German adverb *sicher* ‘certainly’. In contrast, the adjective *sicher* ‘certain’ results in a commitment strengthening (cf. Ernst 2009 for the distinction between subjective and objective epistemic modifiers). This corresponds to findings that in English, epistemic *must* and *certainly* make an assertion defensible under a wider range of circumstances (cf. Lassiter 2016, Yatsushiro et al. 2021).

While the communicative role of epistemic operators is well-established for assertions, their function in questions is less clear. That they do occur in questions can be easily verified. For English, this is shown by Hacquard & Wellwood (2012) in a corpus analysis; they find that epistemic modal verbs occur in questions, but more rarely than in assertions, and that that epistemic *might* occurs significantly more often in polar

questions than *must*. They explain the comparative rarity by assuming that these operators express the speaker's epistemic state, and that it may be strange that the speaker asks about his or her own epistemic state.

However, there is a mechanism that would support the use of epistemics in questions: the so-called “interrogative flip” of subjective operators. This was observed with grammatical evidentials in Tibetan (Garret 2000) and Cuzco Quechua (Faller 2002) and later for a variety of other languages (e.g., Aikhenvald 2003 for Tariana, Matthewson et al. 2007 for St'át'imcets, Murray 2010 for Cheyenne, Lim 2011 for Korean; see recently Brown 2025 for Sm'algyax). It was also observed for lexical evidentials, like particles (e.g. Zimmermann 2004 for German, Schwager 2010 for Tagalog). Interrogative flip was identified as a more general phenomenon that is relevant for other perspectival operators as well, such as personal taste predicates (cf. Tenny 2006, who coined the term “interrogative flip”, for Japanese). This was modelled as a dependence on a syntactically present perspectival center (Speas and Tenny 2003), or a judge parameter (McCready 2007) that undergoes a shift from speaker to addressee in questions. This explains why other context parameters, like the interpretation of first and second person pronouns, are not shifted in questions. It should be noted that there are also cases of reportative evidentials that are anchored to another perspective holder in assertions and in questions (cf. Aikhenvald 2018 and Faller 2025 for Quechua).

Korotkova (2016), discussing a larger set of languages, proposes that “evidentials always are anchored to the addressee in information-seeking questions” (p. 291), arguing against possible counterexamples brought forward in San Roque et al. (2017) (cf. also Aikhenvald 2004, 2018 for crosslinguistic differences). Korotkova motivates this shift of anchor from the general pragmatic conditions under which information-seeking questions are asked, but points out that questions in which the speaker shows a bias towards one answer allow for evidentials that remain anchored to the speaker (p. 326f.). Bhadra (2020) shows that there are evidential particles in Bangla and other languages that remain speaker-centered in questions. As these particles create a speaker bias, this is not against Korotkova's claim (cf. also Frana & Rawlins 2025 for evidentials and biased questions). But it supports Korotkova's remark that “it would be especially interesting to look at evidentials in various types of biased questions.”

I take this advice but return to the domain of subjective epistemic operators, and will investigate them in assertions, non-biased polar questions, and one type of biased question, namely declarative questions, also known as “rising declaratives”. Epistemic operators are different from evidential operators, as they may have an objective reading that is not specifically anchored to the speaker (cf. Lyons 1977, Ernst 2009). However, I focus here on subjective epistemics that are dependent on the speaker in unembedded declarative sentences in the indicative mood in the language under consideration, which is German. They can be differentiated from objective epistemics, as they do not occur in the scope of negation (cf. e.g. the objective epistemic adjective in *it is not certain that she will come* v and the subjective epistemic adverb in **she will not certainly come*). Subjective epistemics should behave similar to evidentials, as they are dependent on a perspective, which is the speaker in root declarative sentences.

More specifically, I will investigate whether the acceptability of epistemic operators differs in assertions, in polar questions, and in declarative questions, uttered in a particular context. To my knowledge, this has not been done before. This procedure has the potential of not only revealing the availability of the interrogative flip but also tell us something about the pragmatic conditions under which the flip can happen. In

particular, I will investigate the differential behavior of strong vs. weak epistemic operators in German, concentrating on the following expressions:

- (1) strong: *sicherlich* ‘certainly’, *wahrscheinlich* ‘probably’, *wohl* ‘presumably’,
müsste ‘should’
 weak: *vielleicht* ‘perhaps’, *möglicherweise* ‘possibly’, *könnte* ‘could’

The epistemic particles in (1) can all occur in assertions in appropriate contexts, and they are all in common use (all range in the domain of 5 out of 7 points on the DWDS corpus). The English translations are not exact but give a fair representation of the meaning of these particles. Their syntactic form is quite diverse: *sicherlich*, *wahrscheinlich*, *möglicherweise* and *vielleicht* are adverbs, *wohl* is a particle (it cannot occur in the prefield of German sentences), and *müsste* and *könnte* are modal verbs, formally subjunctive, that are restricted to epistemic uses, different from their indicative counterparts *muss* ‘must’ and *kann* ‘can’, which also have other modal uses. All operators are subjective, as they cannot occur in the scope of negation. This holds clearly for the adverbs and the particles, but also for the modal verbs. For example, *könnte* differs from *kann* insofar it lacks the prominent reading of the latter with a wide-scope negation.

- (2) a. *Das kann kein echtes Gold sein.*
 ‘This cannot be true gold’,
 less prominent: ‘It is possible that this isn’t true gold’
 b. *Das könnte kein echtes Gold sein.*
 only: ‘It is possible that this isn’t true gold’

The investigation of strong vs. weak modals in polar questions and in biased declarative questions is motivated by the intuition that their acceptability differs in systematic ways. In particular, according to my own judgment, weak subjective epistemic operators are better in polar questions (PQU) than strong ones (Hypothesis 1). This is illustrated in (3) with the weak operator *vielleicht* and the strong operator *sicherlich*.

- (3) [When looking at a photograph in a friend’s apartment:] PQU
Ist das vielleicht ein Falke? better than *Ist das sicherlich ein Falke?*
 ‘Is this possibly a falcon?’ ‘Is this certainly a falcon?’

Furthermore, according to my judgment, strong subjective epistemic operators are better in declarative questions (DQU) than weak ones (Hypothesis 2). This is illustrated in **Error! Reference source not found.**, again with *vielleicht* and *sicherlich*.

- (4) [When looking at a photograph in a friend’s apartment:] DQU
Das ist sicherlich ein Falke? better than *Das ist vielleicht ein Falke?*
 ‘This is certainly a falcon?’ ‘This is possibly a falcon?’

The two hypotheses are given explicitly in (5).

- (5) **H1:** In PQUs,
 weak epistemic operators occur more naturally than strong ones.
H2: In DQUs,
 strong epistemic operators occur more naturally than weak ones.

As the judgments are subtle, the hypotheses need empirical support. This could come from naturally occurring corpus data (as in Hacquard & Wellwood 2012), who already observed that in questions, weak *might* occurs more frequently than strong *must*. However, the hypotheses can be checked more directly with the help of an experiment, which allows for a tighter control of the utterance context.

One should keep in mind that these hypotheses are about the pragmatic conditions under which strong vs. weak subjective epistemic operators occur naturally. Hence we should expect a broader spread of ratings than in grammaticality judgement tasks. But if supported for strong and weak epistemic operators in general, the two hypotheses can reveal aspects about the pragmatics of interrogative flips in general, as I will argue in the Discussion section.

2 Materials and Methods

The two hypotheses H1 and H2 in (5) were tested in a rating experiment. Participants had the task to read a vignette that asked them to imagine being in a particular communicative situation, which resulted in a prompt for a communicative action towards an addressee. They were then asked to rate three proposed utterances – an assertion, a polar question, and a declarative question, all with the same core proposition – according to their naturalness in the situation on a scale from ★ (very unnatural) to ★★★★★ (completely natural). The situation was designed in such a way that the speaker had a reason to ask a question (that is, the proposition was not known to be true to the addressee), and that both a polar question and a declarative question were appropriate. As the participants had to judge the appropriateness of three sentences (an assertion, a polar question, and a declarative question), they were implicitly invited to compare the appropriateness of the three sentences with each other when giving their rating, which should bring about a greater differentiation between the sentence types.

Altogether, eight vignettes were produced that followed this format. The three sentences were varied so that they included no epistemic modification (null) or one of the seven epistemic modifications listed in (1). Each participant was confronted with each vignette once, and the list for each participant contained all eight epistemic particles. The order of the particles within each list was randomized, and the order of the vignettes were randomized as well. Hence, each participant had to rate eight vignettes, with three sentences each.

In German, polar questions are differentiated from declarative assertions in syntax, as they show sentence-initial verb position. Declarative questions show the same word order as declarative assertions; however, they show rising prosody, and carry a question mark in writing. As this difference might easily be overlooked, the participants were made aware of it in the instruction. They were also asked to pronounce (“nachsprechen”) the utterances for the situation, thus evoking a realization with prosody. See (6) for the instruction in English (the original was given in German).

(6) Instruction (English translation):

Welcome to a short survey about how you use language. Below you will find individual scenes that you should imagine yourself in. Each scene presents three possible verbal responses. These are statements or questions—please pay attention to the punctuation! Please read the description of the situation carefully and pronounce the three possible utterances as naturally as possible for that situation. Indicate how natural you find the utterance in each situation. There are a total of eight such situations; the whole thing should take no longer than ten minutes. Let's begin!

One vignette with the adverb *vielleicht* ‘perhaps’ is illustrated in (7). The three sentences were always presented in this order (assertion, declarative question, polar

question), they were highlighted by boldface, and all three sentences and the vignette remained visible during the rating. The participant had to click on one of the stars, which made this star and the stars to its left go black, following the popular rating system on the internet for goods and services. The participants could change their rating, and even go back to previous vignettes and change their rating. This was not encouraged and, in fact, not mentioned. All eight items had to be rated before participants could collect their reward.

- (7) Sie sind zu Besuch auf einer Geburtstagsparty, die Ihre Chefin bei sich zuhause gibt. Sie sind der erste Gast und sehen sich im Wohnzimmer ein wenig um. Auf einem Regal stehen einige Fotos. Sie betrachten eines, das einen etwas älteren Mann mit Anzug und Krawatte zeigt. Sie sagen zu Ihrer Chefin:
[Bitte bewerten Sie die Äußerung mit 1 Stern (sehr unnatürlich) bis 5 Sterne (völlig natürlich)]
- a. **“Das ist vielleicht Ihr Vater.”**
Wie natürlich ist diese Äußerungen auf einer Skala von 1 bis 5?
Sehr unnatürlich ☆ ☆ ☆ ☆ ☆ Ganz natürlich
 - b. **“Das ist vielleicht Ihr Vater?”**
Wie natürlich ist diese Äußerungen auf einer Skala von 1 bis 5?
Sehr unnatürlich ☆ ☆ ☆ ☆ ☆ Ganz natürlich
 - c. **“Ist das vielleicht Ihr Vater?”**
Wie natürlich ist diese Äußerungen auf einer Skala von 1 bis 5?
Sehr unnatürlich ☆ ☆ ☆ ☆ ☆ Ganz natürlich

(8) gives an English translation for (7). One should be reminded that *perhaps* is not a perfect equivalent of *vielleicht*. See appendix for the other seven vignettes, and examples for all modifications of the sentences

- (8) You're visiting a birthday party your boss is hosting at her home. You're the first guest and take a look around the living room. There are some photos on a shelf. You look at one that shows a somewhat older man in a suit and tie. You say to your boss:
[Please rate the statement from 1 star (very unnatural) to 5 stars (completely natural)]
- a. **“That is perhaps your father.”**
How natural is this statement on a scale of 1 to 5?
Very unnatural ☆ ☆ ☆ ☆ ☆ Completely natural
 - b. **“That is perhaps your father?”**
How natural is this statement on a scale of 1 to 5?
Very unnatural ☆ ☆ ☆ ☆ ☆ Completely natural
 - c. **“Is that perhaps your father?”**
How natural is this statement on a scale of 1 to 5?
Very unnatural ☆ ☆ ☆ ☆ ☆ Completely natural

The experiment was conducted online. I recruited 280 participants via Clickworker, with the requirement of native competence of German, which is checked by this portal. The participants were forwarded to a Microsoft Office form, where they were introduced into the experiment, and could perform the experiment. I constructed 8 lists, each participant saw every vignette once, and every treatment of the three sentences once, in a Latin Square design. The first vignette in each list presented the sentence without modification; the subsequent modifications were presented in randomized

order, different for every list. This should ensure that participants started out with a baseline, the unmodified sentences. Each list was completed by at least 34 participants. The experiment was carried out in two phases in July 2024 and December 2024. The lists were made available spaced out at least a day, in order to minimize that participants took part multiple times. Participants were rewarded by 1.50 Euros for their effort, which took them a median time of 4 minutes and 5 seconds. The experiment was covered by the ethics approval of the ERC SPAGAD project.

There was no exclusion of participants. I actually provided an exclusion criterion with a final vignette, which mentioned white wine, but the sentence contained “red wine” instead. I expected a bad rating by attentive participants in all conditions. However, this mistake was detected rarely, or was considered to be an error of the design of the experiment, as indicated in some comments in the comment line that participants could fill out after the experiment. For this reason I decided to drop the exclusion criterion.

3 Results

I present first the ratings for non-modified sentences, for sentences modified by strong epistemic operators, and for sentences modified by weak epistemic operators. Table 1 gives the overall results of the experiment, where strong and weak operators are grouped together. Table 2 shows the percentages of answers.

Rating	Null			Strong			Weak		
	ASS	DQU	PQU	ASS	DQU	PQU	ASS	DQU	PQU
1 ★	83	58	4	88	174	442	122	239	20
2 ★★	51	70	6	138	212	238	125	206	27
3 ★★★	48	84	11	205	263	179	177	230	73
4 ★★★★	32	44	30	267	233	121	163	110	156
5 ★★★★★	67	25	230	425	241	141	255	57	566
Total	281	281	281	1123	1123	1121	842	842	842
Average	2,8	2,7	4,7	3,7	3,1	2,4	3,4	2,5	4,5
StDev	1,6	1,2	0,8	1,3	1,4	1,4	1,4	1,2	0,9
Median	3	3	5	4	3	2	3	2	5

Table 1: Results of rating experiment for assertions, declarative questions and polar questions, for sentences with no epistemic modification, with strong epistemic operators (*sicherlich, wohl, wahrscheinlich, müsste*) and with weak epistemic operators (*vielleicht, möglicherweise, könnte*).

Rating	Null			Strong			Weak		
	ASS	DQU	PQU	ASS	DQU	PQU	ASS	DQU	PQU
1 ★	30%	21%	1%	8%	15%	39%	14%	28%	2%
2 ★★	18%	25%	2%	12%	19%	21%	15%	24%	3%
3 ★★★	17%	30%	4%	18%	23%	16%	21%	27%	9%
4 ★★★★	11%	16%	11%	24%	21%	11%	19%	13%	19%
5 ★★★★★	24%	9%	82%	38%	21%	13%	30%	7%	67%

Table 2: Ratings of Assertions, declarative questions, and polar questions for null, strong, and weak epistemic modification (rounded values)

The overall results are represented in Figure 3. We see with the varied judgments for assertions of unmodified (null) sentences (ASS+Null), compared to the very good judgment for questions of null sentences (PQU+Null), that the vignettes worked: They evoked a situations in which the speaker tends to be ignorant about the proposition of the test sentence. I had expected lower ratings for assertions in this case.

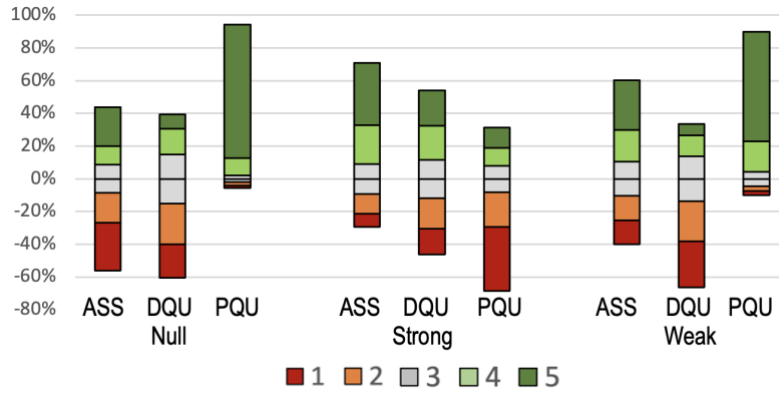


Figure 1: Ratings of Assertions, declarative questions, and polar questions for null, strong, and weak epistemic modification.

Hypothesis **H1** is born out: Weak operators in polar questions are good; in fact, they are nearly as good as null-marked questions. In contrast, strong operators in polar questions reduce the rating, compared to polar questions with null operator (Mann-Whitney Test between PQU+Strong and PQU+Weak, $p < 0.0001$).

Hypothesis **H2** is born out as well: In declarative questions, strong operators are ranked better than weak operators (Mann-Whitney Test between DQU Strong and DQU Weak, $p < 0.001$). However, this difference is not as pronounced as with Hypothesis H1. As for H1, the means are 2 for strong operators vs. 5 for weak operators (averages 2.4 and 4.5); for H2, the means are 3 for strong operators and 2 for mean operators (averages 3.1 and 2.5) (see Table 1).

The differential data for all operators is presented in Figure 2.

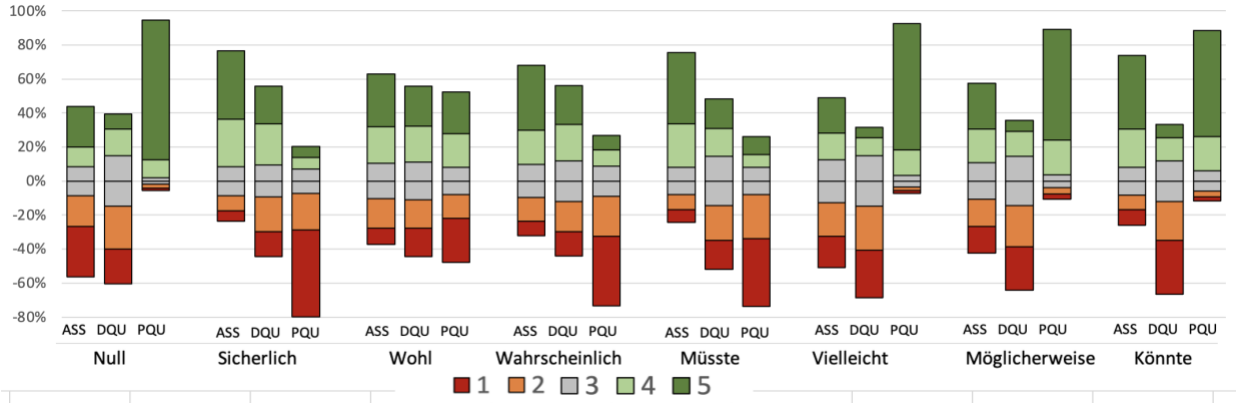


Figure 2: Ratings of Assertions, declarative questions, and polar questions for null modification und for modification with the strong operators *sicherlich*, *wohl*, *wahrscheinlich*, *müsste* and the weak operators *vielleicht*, *möglicherweise* and *könnte*.

We observe that strong operators generally lower the rating of polar questions (compare PQU null with PQU+*Sicherlich*, *Wohl*, *Wahrscheinlich* and *Müsste*). Also, for strong operators, DQUs are better PQUs, except for *Wohl*, for which the difference is not significant (the difference between ASS+*Wohl* and PQU+*Wohl* is close to significant with $p=0.056$).

The two hypotheses are supported for each pair of strong and weak epistemic modifiers. For example, PQU+*Wohl* (the best-rated case of PQU+Strong) is rated significantly below each case of PQU+Weak operator. And DQU+*Müsste* (the worst-rated case of DQU+Strong) is rated significantly above each case of DQU+Weak.

Epistemic operators in assertions generally increase the rating in assertions. This is to be expected, as in the vignettes, the speaker is characterized as not having sufficient information to utter an assertion, and the subjective epistemic operators mitigate the truth commitment of the speaker. The difference is not pronounced for *vielleicht* (cf. ASS+null vs. ASS+*vielleicht*, not significant with $p=0.1$); it is significant for *möglicherweise* ($p<0.001$).

4 Discussion

We set out to investigate the issue whether subjective epistemic operators occur in questions. The motivation for this was that the role of epistemic operators in assertions is clear, namely, to mitigate the speaker's truth commitment, but their role in questions is less obvious. Considering the discussion of so-called interrogative flip for evidential operators, we predicted that there could be a similar flip of epistemic operators, which then would mitigate the truth commitment of the addressee when answering the question. However, as Hacquard & Wellwood (2012) observe, it is unclear why the speaker would ask the addressee to utter an assertion modified by a subjective epistemic.

Given the discussion of the presence or absence of interrogative flip for evidential operators, we considered two types of questions, namely regular information-seeking polar questions (PQUs) on the one hand, and declarative questions (DQUs) on the other. The latter have a bias of the speaker towards the truth of the proposition, hence are somewhat between assertions and polar questions. In looking for experimental

evidence, we formulated two hypotheses based on introspection that took account the nature of the epistemic operators: While epistemic operators are less accepted in polar questions than in assertions, this only holds for strong epistemic operators, not for weak ones (H1). And while this may hold in polar questions, things may be the other way round for declarative questions, for which strong epistemic modifiers appear to be better than weak ones (H2). If corroborated, these tendencies should provide for clues of the problem why speakers would encourage the use of subjective epistemic modifiers in the answers to questions.

The experiment, carried out for German, found strong support for both hypotheses. Weak subjective epistemic modifiers are preferred over strong ones in polar questions, and strong subjective epistemic modifiers are preferred over weak ones in declarative questions. This holds for all the four strong and three weak epistemic operators selected for the experiment individually.

4.1 Hypothesis 1: Weak epistemic operators in polar questions

We turn now to the rationale behind the first hypothesis. Why do strong subjective epistemic operators reduce ratings in polar questions, but not weak ones? For the first observation concerning strong subjective operators, notice that with a standard polar question, the speaker requests information from the addressee that resolves the issue whether a proposition under discussion is true. The speaker expects that optimally, the addressee should provide a truth commitment of default strength towards the proposition or towards its negation. Epistemic operators, even “strong” ones like *must*, reduce the strength of an assertion, by making the commitment easier to defend in case it turns out to be false (Yatsushiro et al. 2021, Krifka 2026). While the addressee might actually choose to answer a question under epistemic weakening, this is not in the interest of the speaker when asking the question. This explains why subjective epistemic modifiers should not occur in questions.

Note that this reasoning does not apply to objective epistemic operators. It is perfectly possible to ask, for example, *Is it certain that the train will arrive on time?* The reason is that objective epistemic qualifications can be part of the proposition that answers the question under discussion. This also holds for subjective epistemic states that are expressed by operators like adjectives, as in *Are you certain that the train will arrive on time?* or *Do you know whether the train will arrive on time?* In contrast to such cases, the subjective epistemic operators we investigated are systematically not part of the question under discussion.

But why, then, are subjective modifiers good in polar questions when they are weak? Given the reasoning just outlined, they should be even worse, because the speaker allows the addressee to answer a question under even lower truth commitment. But this is precisely the point: If the speaker reckons that the addressee might have insufficient information, or wants to be polite by not putting pressure on the addressee, the speaker may grant epistemic operators in the question that the addressee can use in the answer (cf. Bhadra 2020 for evidentials). For example, if the speaker asks *Is it possibly raining?* the speaker signals to the addressee that assertions like *It is possibly raining* would be acceptable answers, flipping the subjective epistemic to the addressee. In this pragmatic constellation, when the speaker assumes that the addressee may not have sufficient knowledge but wants information from the addressee anyway, we expect the speaker to suggest to the addressee that the truth commitment in the answer can be mitigated, and in particular, that weak epistemic operator can be used.

The motivation for this pragmatic move is related to the use of negative polarity items like *any* and *ever* that broaden the proposition in polar questions (Krifka 1995, van Rooij 2004), as in *Did you see any suspicious person?* or *Did you ever smoke marihuana?* There, the speaker semantically weakens the proposition in order to increase the chance for a positive answer, as this increases the informativity of the question. Technically, the entropy of the question is increased, as the likelihood of each proposition is more similar than without the modification. In a similar way, allowing for answers with weak subjective modifiers increases the chances for a confirming answer even if the addressee does not have full commands over the facts, and hence increases the potential informational gain that the speaker gets out of the answer.

I would like to add that while polar questions do not favor strong subjective modifiers that lead to a slight mitigation, in contrast to those that lead to a more severe mitigation, they apparently can be strengthened by the speaker to increase the requested truth commitment from the addressee. This is the case with information that the addressee otherwise would not release, or would not express in a clear way. Examples are *Is this really your father?* or German *Ist das wirklich Ihr Vater?* Expressions like *really*, *definitely* and German *wirklich*, *tatsächlich* should not be analyzed as strong epistemics, but rather as commitment raisers (cf. Krifka 2023). Hence, interestingly, polar questions allow for raising the truth commitments for the addressee, or for lowering it considerably, in different pragmatic constellations. What is pragmatically odd is to lower them just a little bit. The option for commitment raisers in polar questions would have to be experimentally investigated.

4.2 Hypothesis 2: Strong epistemic operators in declarative questions

We now turn to the rationale behind the second hypothesis, H2. Why do subjective epistemic modifiers occur in declarative questions, and why are, in particular, strong subjective epistemic modifiers rated as very good in such sentences? For this, we have to consider the pragmatics of declarative questions. DQUs differ from PQUs as they show an epistemic bias of the speaker towards the core proposition. The origin of this bias can be explained as follows (cf. Gunlogson 2008, Malamud & Stephenson 2015): In a PQU, the speaker checks whether the addressee commits to the proposition, whereas in a DQU, it is the speaker who proposes to commit to the proposition, but makes this commitment dependent on whether the addressee commits to it as well. Rising declaratives signal a “contingent” (Gunlogson 2008), “conditional” (Farkas & Roelofsen 2017) or “tentative” commitment (Bhadra 2020) by the speaker. Hence, subjective epistemics are interpreted with respect to the speaker, and are not flipped to the addressee. The typical pragmatic constellation in which a DQU is asked is when the speaker has rather good (but not perfect) evidence for the proposition. Consequently, we should expect that DQUs naturally combine with strong epistemic operators, which signal this good but not perfect evidence. The evidence expressed by weak modifiers is not strong enough to reach the level of evidence that is required by declarative questions.

I motivated the distribution of subjective epistemic operators in polar questions and declarative questions by arguing that in the former, they flip to the addressee, which favors weak epistemic operators, and in the latter, they remain with the speaker, which favors strong epistemic operators. This corresponds to the representation of these question types in Krifka (2021, 2023), where I proposed that in a polar question, the speaker asks for a commitment of the addressee to the proposition, whereas in a

declarative question, it is the speaker who proposes a commitment to the proposition, provided that the addressee does not object to it. This is mediated by a judge parameter, which is shifted to the addressee in polar questions but remain anchored to the speaker in declarative questions. Subjective epistemic operators are interpreted with respect to the judge parameter, and their interpretation (flipped to the addressee in polar questions, remaining with the speaker in declarative questions) is a consequence of this modelling.

It should be noted that this reasoning is valid only for declarative questions in which the speaker seeks support from the addressee. There is another type that is marked by a distinct prosody that expresses incredulity, cf. Cohen (2007). Goodhue (2021) distinguishes between “confirmational” and “incredulous” rising declaratives; in the experiment, we just dealt with the first type, not with the second. In incredulous declarative questions, subjective epistemic modifiers do not occur. For example, when being told by a person that a particular photograph shows a picture of the speaker’s father even though he looks very different from the speaker, one can react by *This is your father?!*, with incredulity prosody (cf. Repp & Seeliger 2023 for the prosodic realization of incredulity in German). The use of subjective epistemics is ruled out: *This is (certainly / possibly) your father?! is impossible*. The declarative sentence with an incredulity contour is not a tentative assertion that needs approval by the addressee – quite the opposite, it is a challenge to an utterance of the addressee

There is evidence that incredulous declaratives are just a subcase of incredulity responses (I acknowledge discussions with Ariel Cohen about this topic). In such responses, the speaker expresses incredulity about a phenomenon in the situation, for example, a speech act of the other speaker. This can be an assertion (like *This is my father. – This is your father?! That’s unbelievable!*), a question (like *What is the capital of Paris? – What is the capital of Paris?! Everyone knows that!*), but also a command or even an exclamation. If the initial speech act contained an epistemic operator, it can also incur in the incredulity response, as in *This is possibly my father. – This is possibly your father?! You don’t know for sure?* But epistemic operators cannot be added to the material of the antecedent utterance that the speaker reacts to.

4.3 What’s up with *wohl* ?

In our survey of different subjective epistemic operators, we found that *wohl* sticks out. This discourse particle is significantly better in PQUs than other strong epistemic operators (comparison with PQU+*Wahrscheinlich*, $p < 0.001$). In fact, this is the only particle which is rated similarly in PQUs and DQUs, as the difference between these two cases is not significant. At the same time, it is evident that *wohl* patterns with the strong epistemic modals – the ratings of PQU+*Wohl* are much worse, and the ratings of DQU+*Wohl* much better, than for weak quantifiers.

The particle *wohl* has attracted a lot of attention in the literature on discourse particles. In particular, Zimmermann (2004, 2011) argues that *wohl* is dependent on the sentence type it occurs in, and is anchored to the speaker in assertions and to the addressee in polar questions. However, at the same time, *wohl* clearly belongs to the strong epistemic operators, and following the reasoning of Section 4.1, the acceptability in polar questions should be decreased. But there is disagreement in the literature: According to Doherty (1985: 81), *wohl* does not occur in polar questions but does occur in declarative questions (her examples and judgements are **Ist Konrad wohl verreist?* ‘Is Konrad WOHL away travelling?’ and *Konrad ist wohl verreist?*). I share this judgment, at least as a tendency. Hence, there is the possibility for speaker or dialectal differences in the

understanding of *wohl* that would be worthwhile to investigate further. Also, PQU+*Wohl* exhibit the greatest standard variation, or participant disagreement, in the ratings, after ASS+null. Despite the great interest in the particle *wohl*, the issue of speaker variation has not been addressed so far. Also possible differences of *wohl* in polar questions vs. declarative questions have not been studied. For example, Repp and Seeliger (2025) addresses the prosodic realization of *wohl* in assertions and declarative questions, but leaves polar questions aside.

The particle *wohl* also occurs in constituent questions, both with verb-second position as in *Wo ist wohl der Schlüssel?* ‘where is WOHL the key’ and with verb-final position as in *Wo wohl der Schlüssel ist?* In the latter case, it is obligatory to form a question, and it marks conjectural questions that raise an issue but are not necessarily expected to be answered (Truckenbrodt 2006). Interestingly, the other epistemic modifiers discussed in this article do not occur easily in constituent questions, with the exception of *könnte* (*Wo könnte der Schlüssel sein?*) – a topic for future research.

One reason why *wohl* behaves differently from modifiers like *sicherlich* und *müsste* could be that it is not an epistemic operator but rather an inferential evidential. This is argued for by Göbel (2018), who also gives data that *wohl* can be used in assertions that the speaker assumes to be true, but for which the nature of evidence is inferential. The evidential interpretation could result in better rating of PQU-*Wohl* items. However, notice that the ratings of PQU-*Wohl* are not particularly good. It might be that an addressee-anchored inferential interpretation was plausible for some participants but not for others. It would be worthwhile to compare the ratings for *wohl* with clear evidentials, like *offenbar* ‘obviously’.

Another line of explanation for the special behavior of *wohl* can be derived from Zimmermann’s observation that this particle is felicitous when both the speaker is uncertain about the proposition and assumes that the addressee may be uncertain about it as well (a configuration that was not investigated in the experiment, as the vignettes tried to describe the addressee as an expert). Zimmermann’s example is one hiker asking a fellow-hiker, *Ist dies wohl der richtige Weg?* ‘Is this WOHL the right way?’ We can hypothesize that the perspective of *wohl* always includes the speaker, as part of its lexical meaning. In polar questions, for which the perspective always includes the addressee, this results in a combined speaker + addressee perspective. Hence for speakers that allow for PQU+*Wohl*, this combination encodes the assumptions that both speaker and addressee may have less than default knowledge about the truth of the proposition. At first sight, this is problematic with another observation by Zimmermann, namely that *wohl* is fine in exam questions in which the speaker has full knowledge, as in a teacher asking a student, *Was ist wohl die Hauptstadt von Papua Neuguinea?* ‘What is WOHL the capital of Papua New Guinea?’ But such cases can be explained as an alignment of the speaker with the addressee that we also observe in cases when a doctor asks a patient, *How are we feeling today?* The scenarios investigated in the experiment were not designed to facilitate this speaker/addressee alignment, but they do not exclude it either. The better rating of PQU+*Wohl*, compared to other PQU+strong cases, might be due to this possibility of an aligned interpretation. This hypothesis could be investigated in an experiment that compares cases that favor the aligned interpretation, and cases that discourage it.

5 Conclusion

This article has provided experimental evidence that interrogative flip indeed exists, for subjective episodic operators. The motivation for this claim comes from the fact that polar questions, which are claimed to be able to show interrogative flip, and declarative questions, for which there are good theoretical reasons that they don't, behave differently with respect to subjective epistemic operators: Polar questions, which exhibit the flip, prefer weak epistemic operators, whereas declarative questions, which do not exhibit flip, prefer strong epistemic operators. We have motivated this by general pragmatic considerations about possible motivations to have subjective epistemic operators in polar questions and in declarative questions.

One important lesson for future research is that it is necessary to consider a wider variety of subjective operators when investigating perspectival shift phenomena. It is my impression that previous research was limited to too few cases, like the discourse particle *wohl* in German. Interestingly, it turned out that *wohl* may be special among the strong subjective epistemic operators.

There are a number of factors that were not part of the study. One is that it worked with written prompts, and could not investigate the effects of prosody in detail, beyond their representation by punctuation (see Repp & Seeliger 2025 for a production study devoted to this issue). Another point is that it did not investigate possible, and likely, variations between speakers that may or may not be based on dialectal differences. Third, the study investigated only a small number of subjective epistemic operators. There are others, and there other types of operators – objective epistemic operators, various types of evidentials, and strengtheners of commitments like *definitely*. Hopefully, future studies will reveal more about their function in interactive communication.

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The Excel file with the data can be downloaded here [Epistemic Questions Data Krifka.xlsx](#) and under <https://tinyurl.com/Epistemics-ASS-DQU-PQU>

Appendix

Seven vignettes, in English translation (Google translate), with non-modified sentences.

- (9) You are a guest at an Italian family restaurant that boasts that famous personalities have dined there. Many framed photographs from past decades are displayed on one wall. You look at one picture with particular interest because a person in it seems somehow familiar. You say to the waiter:
a. "That's Sophia Loren." b. "That's Sophia Loren?" c. "Is that Sophia Loren?"
- (10) You're visiting a good friend from school who has been living in Munich for several years now. She's a passionate art lover, and naturally, she's showing you

the city's famous museums. You yourself aren't nearly as familiar with paintings as your friend. You're walking together through the halls of the Alte Pinakothek and heading towards a small portrait. You say to your friend:

a. "That's a Rembrandt." b. "That's a Rembrandt?" c. "Is that a Rembrandt?"

- (11) Your colleague shows you pictures from his recent vacation in Greece during your lunch break. He raves about the Greek islands he visited with his family. You yourself have never been to Greece, but of course you know about their beauty. Your colleague is showing you a picture of his two children in front of a small Orthodox chapel with a blue dome, perched on a rock above the sea. You say to your colleague:

a. "That was on Santorini." b. "That was on Santorini?" c. "Was that on Santorini?"

- (12) You've enrolled in an adult education course on ornithology because you finally want to learn how to identify birds – you have no experience with this. You're currently on your first field trip in a moorland area and well-equipped with binoculars. The instructor points out a bird in the top of a birch tree. You say to the instructor:

a. "That's a lapwing." b. "That's a lapwing?" c. "Is that a lapwing?"

- (13) You're walking down the street with your old friend to a beer garden this evening. Your friend is a car enthusiast, while you're not very knowledgeable about cars. A little ways off, you spot a striking sports car at a red light and point it out to your friend. You say to your friend:

a. "That's an Alfa Romeo." b. "That's an Alfa Romeo?" c. "Is that an Alfa Romeo?"

- (14) In German class, the teacher handed out poems without naming the authors. The exercise involves identifying the authors of the poems based on their style. You are quite unsure about one poem and ask the teacher. You say to your teacher:

a. "That's a poem by Heine." b. "That's a poem by Heine?" c. "Is that a poem by Heine?"

- (15) Your neighbor is a passionate mountaineer. She has already completed many tours in the Alps. You, on the other hand, are not familiar with the Alps. Your neighbor tries to share her enthusiasm with you. You are both looking at a beautiful book of mountain photographs from Austria. You turn a page and see a particularly beautiful peak. You say to your neighbor:

a. "That's the Großglockner." b. "That's the Großglockner?" c. "Is that the Großglockner?"

Variations of a sentence:

- (16) a. Das ist Ihr Vater. / Das ist Ihr Vater? / Ist das Ihr Vater?
b. Das ist sicherlich Ihr Vater. / Das ist sicherlich Ihr Vater? / Ist das sicherlich Ihr Vater?
c. Das ist wohl Ihr Vater. / Das ist wohl Ihr Vater? / Ist das wohl Ihr Vater?
d. Das ist wahrscheinlich Ihr Vater. / Das ist wahrscheinlich Ihr Vater? / Ist das wahrscheinlich Ihr Vater?
e. Das ist möglicherweise Ihr Vater. / Das ist möglicherweise Ihr Vater? / Ist

- das möglicherweise Ihr Vater?
- f. Ist das vielleicht Ihr Vater? / Das ist vielleicht Ihr Vater? / Ist das vielleicht Ihr Vater?
- g. Das müsste Ihr Vater sein. / Das müsste Ihr Vater sein? / Müsste das Ihr Vater sein?
- h. Das könnte Ihr Vater sein. / Das könnte Ihr Vater sein? / Könnte das Ihr Vater sein?

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